

Data Cleaning

```
library(tidyverse)
```

```
— Attaching core tidyverse packages — tidyverse 2.0.0
—
✓ dplyr      1.1.4    ✓ readr      2.1.5
✓ forcats    1.0.1    ✓ stringr    1.5.2
✓ ggplot2    4.0.0    ✓ tibble     3.3.0
✓ lubridate  1.9.4    ✓ tidyr      1.3.1
✓ purrr      1.1.0
— Conflicts — tidyverse_conflicts()
—
* dplyr::filter() masks stats::filter()
* dplyr::lag()     masks stats::lag()
i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all
conflicts to become errors
```

```
library(readr)
library(sf)
```

Linking to GEOS 3.13.1, GDAL 3.11.0, PROJ 9.6.0; sf_use_s2() is TRUE

```
library(tigris)
```

To enable caching of data, set `options(tigris\_use\_cache = TRUE)`
in your R script or .Rprofile.

```
library(dplyr)
library(ggplot2)
library(stats)

precinct_2024_general_election <- read.csv("data/g24_sov_by_g24_svprec.csv")
```

```
simpler_precinct <- precinct_2024_general_election |>
  select("COUNTY", "TOTVOTE", "CDDIST", "ASSDEM01", "ASSDEM02", "ASSREP01",
         "ASSREP02", "CNGDEM01", "CNGDEM02", "CNGREP01", "CNGREP02",
         "PRSDM01", "PRSREP01", "SENDEM01", "SENDEM02",
```

```

"SENREP01", "SENREP02", "USSDEM01", "USSREP01") |>
  filter(
    !if_any(
      .cols = -COUNTY,
      .fns = ~ . == "***"
    ) |>
    mutate(
      across(
        .cols = -COUNTY,
        .fns = ~ as.integer(as.character(.))
      )
    )
  )

```

Second Part

```

votes_sr <- read_csv("data/state_g24_sov_data_by_g24_srprec.csv")

```

Rows: 25245 Columns: 76

— Column specification

Delimiter: ","

chr (49): FIPS, SRPREC, ELECTION, SRPREC_KEY, GEO_TYPE, ASSAIP01, ASSDEM01, ...

dbl (27): COUNTY, ADDIST, CDDIST, SDDIST, BEDIST, TOTREG, DEMREG, REPREG, AI...

i Use `spec()` to retrieve the full column specification for this data.
i Specify the column types or set `show\_col\_types = FALSE` to quiet this message.

```

block_corr <- read_csv("data/state_g24_sr_blk_map.csv")

```

Rows: 421137 Columns: 14

— Column specification

Delimiter: ","

chr (6): BLOCK_KEY, FIPS, SRPREC, SRPREC_KEY, ELECTION, GEO_TYPE

dbl (8): COUNTY, TRACT, BLOCK, BLKREG, SRTOTREG, PCTSRPREC, BLKTOTREG, PCTBLK

i Use `spec()` to retrieve the full column specification for this data.
i Specify the column types or set `show\_col\_types = FALSE` to quiet this message.

```

new_district_map <- read_csv("data/ab604.csv")

```

Rows: 519722 Columns: 2

— Column specification

Delimiter: ","

chr (2): 060650444033050, 25

i Use `spec()` to retrieve the full column specification for this data.
i Specify the column types or set `show\_col\_types = FALSE` to quiet this message.

```
votes_sr <- votes_sr |>
  select(SRPREC_KEY, CNGDEM01, CNGREP01)
block_corr <- block_corr |>
  select(BLOCK_KEY, SRPREC_KEY, PCTSRPREC)
new_district_map <- read_csv(
  "data/ab604.csv",
  col_names = c("BLOCK_KEY", "NEW_CD"),
) |>
mutate(BLOCK_KEY = as.character(BLOCK_KEY)) |>
select(BLOCK_KEY, NEW_CD)
```

Rows: 519723 Columns: 2

— Column specification

Delimiter: ","

chr (2): BLOCK_KEY, NEW_CD

i Use `spec()` to retrieve the full column specification for this data.
i Specify the column types or set `show\_col\_types = FALSE` to quiet this message.

```
allocated_data <- block_corr |>
  left_join(votes_sr, by = "SRPREC_KEY") |>
  mutate(
    CNGDEM01 = as.numeric(CNGDEM01),
    CNGREP01 = as.numeric(CNGREP01),

    PCTSRPREC = as.numeric(PCTSRPREC)
  ) |>

  mutate(
    D_Allocated = CNGDEM01 * (PCTSRPREC / 100),
    R_Allocated = CNGREP01 * (PCTSRPREC / 100)
  )
```

```
Warning: There were 2 warnings in `mutate()`.
The first warning was:
i In argument: `CNGDEM01 = as.numeric(CNGDEM01)`.
Caused by warning:
! NAs introduced by coercion
i Run `dplyr::last_dplyr_warnings()` to see the 1 remaining warning.
```

```
final_results_ab604 <- allocated_data |>
  left_join(new_district_map, by = "BLOCK_KEY") |>
  filter(!is.na(NEW_CD)) |>
  group_by(NEW_CD) |>

  summarise(
    New_D_Votes = sum(D_Allocated, na.rm = TRUE),
    New_R_Votes = sum(R_Allocated, na.rm = TRUE),
    .groups = 'drop'
  ) |>

  mutate(
    New_Total = New_D_Votes + New_R_Votes,
    New_D_Pct = New_D_Votes / New_Total,
    New_Winner = ifelse(New_D_Votes > New_R_Votes, "DEM", "REP")
  )

final_results_ab604
```

```
# A tibble: 52 × 6
  NEW_CD New_D_Votes New_R_Votes New_Total New_D_Pct New_Winner
  <chr>      <dbl>      <dbl>      <dbl>      <dbl> <chr>
1 01         182536.    150547.    333083.     0.548 DEM
2 02         231970.    148401.    380371.     0.610 DEM
3 03         189935.    164638.    354572.     0.536 DEM
4 04         189348.    143688.    333035.     0.569 DEM
5 05         118108.    206911.    325019.     0.363 REP
6 06         171008.    151904.    322912.     0.530 DEM
7 07         189330.    144663.    333992.     0.567 DEM
8 08         206138.     95613.    301751.     0.683 DEM
9 09         150450.    107208.    257658.     0.584 DEM
10 10         241489.    122289.    363778.     0.664 DEM
# i 42 more rows
```